25-11-21

Electrostatics

Thursday

Electrostatic Force

Force of attraction or repulsion b/w two points charges is called "Electrostatic force"

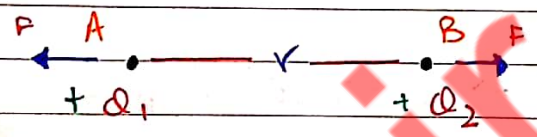
→ charges either repel or attract

Coulomb's Law of Electrostatics

Force of attraction or repulsion b/w two point charges is directly proportional to product of charges and is inversely proportional to square of distance b/w the charges.

↳ **Conditions / Factor's effecting Electrostatic force**

Magnitude of
=> Product of charges => Medium
=> square of distance b/w the charges



* A & B are two point charges
* r is the distance b/w the
* two point charges

$F \propto Q_1 Q_2$ —i

$F \propto \frac{1}{r^2}$ —ii

→ Repulsive forces => +ve
→ Attractive forces => −ve

$F \propto \frac{Q_1 Q_2}{r^2}$

↳ No change if we are representing numerical magnitude

$F = \pm (\text{constant}) \frac{Q_1 Q_2}{r^2}$

↳ ± represents then it can either be attractive or repulsive force

$$\boxed{F = \pm \frac{K Q_1 Q_2}{r^2}}$$ — mathematical expression of coulomb's law of electrostatics

↳ K is called "coulomb's constant"

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Properties of Electrostatic force

1- K = Coloumb's constant

$K = \frac{1}{4\pi\varepsilon}$ ε = epsilon

So $\varepsilon \propto \frac{1}{K}$

* Medium also affects the electrostatic force
↓
different in air & water
↳ property of medium that affects the electrostatic force b/w the charges is called "permitivity of medium"

=> For every medium the permitivity is different

=> Permitivity of air & vacum are almost "same"

=> In our syllabus => only air or vacum

ε = Permitivity of medium

ε_0 = Permitivity of vacum => $8.85 \times 10^{-12} C^2/Nm^2$
↓
given in data booklet

So,

$K = 9 \times 10^9 \, Nm^2/C^2$ (same for vacum & air) * almost

$$F_E = \pm \frac{1}{4\pi\varepsilon_0} \frac{Q_1 Q_2}{r^2}$$

2- -ve sign — attractive force

+ve sign — repulsive force

3- Electrostatic force follows inverse square law

$F \propto \frac{1}{r^2}$

↳ for graph
=> curved graph
=> negative gradient
=> never touches axis
=> more steeper gradient

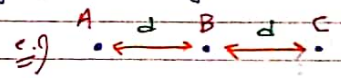
4- Vector Quantity, that is electrostatic force at a point due 2 or more than 2 charges is found by the method of "vector addition"

5- Electrostatic force exists in pair

↳ even if one charge has smaller magnitude and the larger magnitude for other charge both the charges will exert "equal electrostatic force"
⇓
bcz it depends on the product of charges and it remains the same so the F_E remains same.

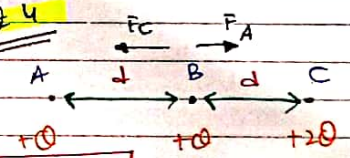
6- Electrostatic force always acts along the line joining the two charges

e.g.



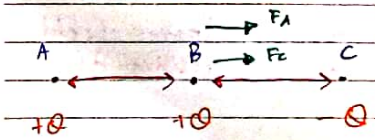
↳ so same along the line

Continuation of point 4



$$F = F_C - F_A$$ — resultant force

* If we have to find resultant force at B we will use vector addition method



$$F = F_A + F_C$$ — resultant force

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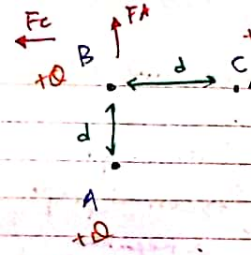


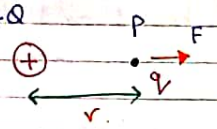
$$F = \sqrt{F_A^2 + F_C^2}$$

Electric field Strength

Force acting on a unit charge at a point inside the electric field is called "Electric field strength at that point"

$$\text{Electric field strength} = \frac{\text{Force}}{\text{charge}}$$

$$E = \frac{F}{q}$$



=> q is a charge introduced in an electric field of "⊕/Q"

positively charged

$$F = \frac{kQq}{r^2}$$

$$E = \frac{F}{q}$$

$$E = \frac{kQ\cancel{q}}{r^2} \div \cancel{q}$$

$$E = \frac{kQ}{r^2}$$

→ This expression is for the electric field strength on a "Unit Positive Charge"

E = Electric field strength on a unit positive charge
k = Coloumb's constant
Q = charge on the point in which's electric field matter
r² = Distance b/w the two charges or "Q and P"

charge "q" is introduced / present

→ Here, as the charges attract each other

$$F = \frac{-kQq}{r^2}$$

→ we don't mention or show -ve sign while noting down the magnitude of the electric field strength.

$$E = \frac{F}{q}$$

$$E = \frac{-kQ}{r^2}$$

→ Electric field strength for a unit negative charge

Properties

1- Unit = $\frac{N}{C}$ or $\frac{V}{m}$ $\left(\frac{\Delta V}{d}\right)$ ← Another expression for electric field strength

2- Follows inverse square law i.e $E \propto \frac{1}{r^2}$

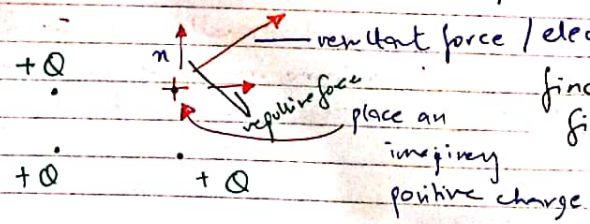
3- Electric field strength is a vector quantity

→ We have to specify it's direction

→ At any point, place an imaginary unit positive charge and identify the force's direction on the charge, the direction of force will be the direction of the electric field strength

→ It's a vector quantity, that is it is an electric field strength at a point due to two or more than two charges is found by the method of "vector Addition"

Q.



find electric field strength on "x"

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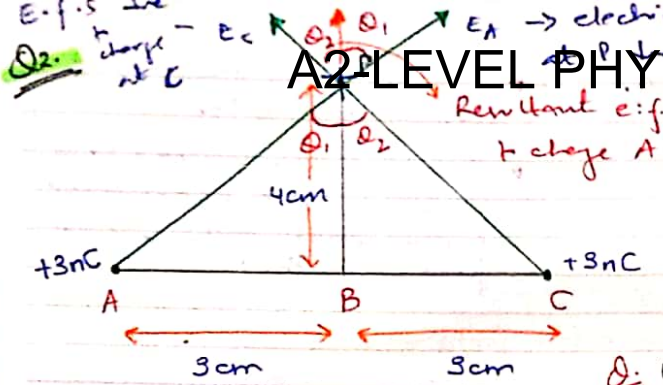
E.f.s due to charge at C — E_C

E_A → electric field strength due to charge at A

Q2.



Resultant e.f.s at P due to charge A & C

Q. find magnitude & direction of electric field strength due to charge at A & C

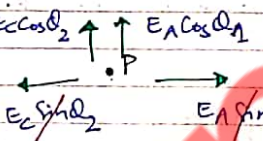
Steps

1. Draw a positive charge at P
2. Determine / identify the electric field strength on unit positive charge — as E_A & E_C
3. $E = \frac{kQ}{r^2}$

   ↳ $E_A = \frac{kQ_1}{r^2}$ & $E_C = \frac{kQ_C}{r^2}$

4. Resolve the two (E_A & E_C) — electric field strength

   ↓ done on diagram ⇒



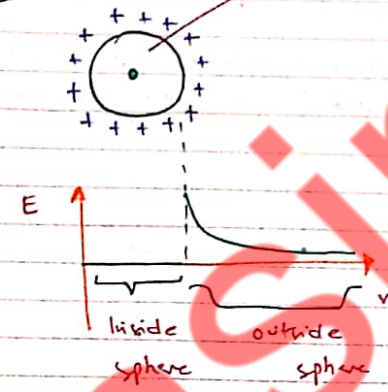
↳ they cancel each other out due to being same magnitude and exactly opposite directions

⇒ same magnitude cz same charge and force acting on it ⇒ same θ also

5. Resultant force / E.f.s acting at P is

$$\boxed{E = E_A\cos\theta_1 + E_C\cos\theta_2}$$

Q. plot graph of "E vs r" inside & outside the sphere



$E = \frac{kQ}{r^2}$

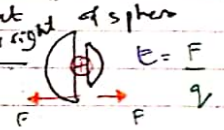
∴ $E \propto \frac{1}{r^2}$

Electric field strength is zero inside the sphere

e.f.s at centre of sphere

$E = \frac{F}{q}$

e.f.s at further right of sphere



$E = \frac{F}{q}$

Same force will be exerted by each portion so resultant force zero thus electric field strength will be zero inside the sphere

⇓

Force will be same cz electric force depends on both charge & distance

e.g. left portion charge increases & distance increases

right portion charge decreases & distance decreases

⇓

thus ratio remains same

Same equal halves so equal charge, so same force acting in opposite directions thus force will zero thus electric field strength will be zero

→ Resultant force per unit charge at all points inside a charged conductor is zero that's why electric field strength at all the points inside charged conductor is zero

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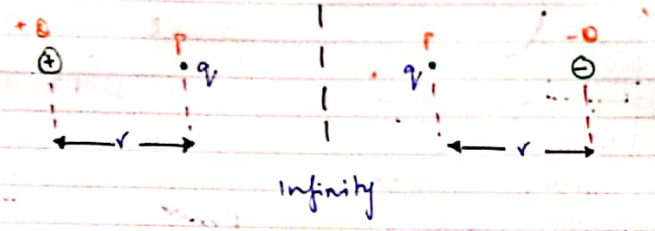
Electric Potential Energy

→ Represents work done

Work done in moving a charge from infinity to a point inside electric field is called "Electric potential energy of the charge at that point"



→ Electric potential of charge "q" is work done in moving charge 'q' from infinity to point P

$E.P.E = \frac{kQq}{r}$ | $E.P.E = -\frac{kQq}{r}$

↳ Between + and - , it's attractive so -ve

Q- Magnitude of charge in whose electric field we have placed charge in.

→ Electric potential energy can be negative and positive

Properties of Electric Potential Energy

1- Unit is "Joule" as it's energy

2- It is a scalar quantity (No direction as energy does not have a direction)

↳ negative sign is part of magnitude in scalar quantity

↳ In vector quantity, negative is not part of magnitude

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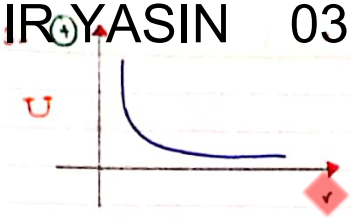


$U \propto \frac{1}{r}$ as k, Q and q are constant

$U = -\frac{kQq}{r}$ / $\frac{kQq}{r}$

Graph Properties

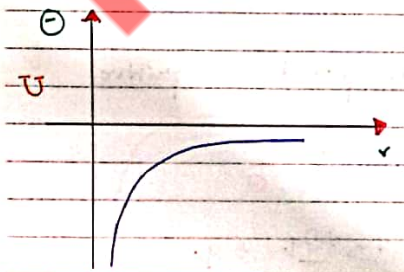
i) Electric potential energy is maximum at surface of positive charge

ii) Electric potential decreases as we move away from positive charge

iii) Electric potential energy is zero at infinity and it's at minimum value

↳ if we place ∞ infinity in place of r in the formula $E.P.E = \frac{kQq}{r}$

Since minimum at infinity hence curve does not touch x-axis



$U \propto -\frac{1}{r}$ for negative charge

i) E.P.E is minimum at surface of negative charge

ii) E.P.E increases as we move away from negative charge

iii) E.P.E is zero at infinity and is at it's maximum value for negative charge

4- Gradient $= \frac{\Delta U}{\Delta r}$ ⇒ $\frac{\text{change in energy}}{\text{change in distance}}$ ⇒ $\frac{U}{r}$

↳ put $\frac{kQq}{r}$ in place of U

⇒ $\frac{kQq}{r} \div r$ so $\frac{kQq}{r^2}$ ∴ Force

→ Gradient of E.P.E vs distance graph represents electric force

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Electric Potential

E.P.E of unit charge at a point inside electric field is called "electric potential at that point"

→ this is electric potential of unit charge at a point

→ E.P.E is electric potential energy while electric potential is different

"OR"

Work done in moving unit charge from infinity to a point inside electric field is called "electric potential" at that charge

→ this defination to written in exam

$$\text{Electric Potential} = \frac{\text{E.P.E}}{\text{charge}}$$

$$\boxed{V = \frac{U}{Q}}$$ → general expression for electric potential

When Q is positive charge, Electric potential in positive charge field;

$U = \frac{kQq}{r}$ since, $V = \frac{U}{Q}$

$V = \frac{kQq}{r} \div q$

$$\boxed{V = \frac{kQ}{r}}$$ → Electric potential in positive charge

Electric potential in negative electric field;

$U = -\frac{kQq}{r}$

$V = \frac{U}{Q}$

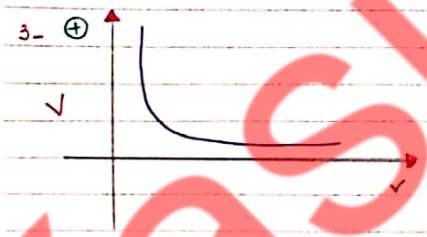
$$\boxed{V = \frac{-kQ}{r}}$$ → Electric Potential for -ve charge

Properties of Electric Potential

1- Unit is $JC^{-1} = V$

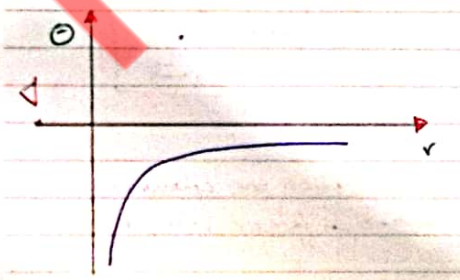
2- Scalar quantity

3- ⊕



i) Electric potential is maximum at surface of positive charge
ii) Electric potential decreases as we move away from positive charge
iii) Electric potential is zero at infinity and thus it's minium value

⊖



i) Electric potential is minimum at surface of ~~positive~~ negative charge
ii) Electric potential increases as we move away from negative charge
iii) Electric potential is zero at infinity and it is at it's minium value for negative charge

4- Gradient $= \frac{\Delta V}{\Delta r}$

$= \frac{V}{r}$

$= \frac{kQ}{r} \div r$

$= \frac{kQ}{r} \times \frac{1}{r}$

$$= \boxed{\frac{kQ}{r^2}}$$ → This represents electric field strength

→ Gradient of electric potential vs distance graph represents "electric field strength"

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Electric field strength → potential vs distance graph (V vs r) graph

$$E = \frac{\Delta V}{\Delta r}$$

↳ Δr is distance (change in distance)

So,

$$\boxed{E = \frac{\Delta V}{d}}$$ formula of Ac

Potential Difference

Work done in moving a charge between two points is called "potential difference" b/w the 2 points.

$$\text{potential difference} = \frac{\text{Work Done}}{\text{charge}}$$

$$\boxed{\Delta V = \frac{W}{q}}$$

$$\boxed{\text{volt} = \frac{\text{Joule}}{\text{coloumb}}}$$

1 Volt

If one joule of work is done in moving one coloumb charge between the two points then potential difference between the two points is one volt

$$\text{Work done} = \Delta V \times q$$

Electron Volt

Work done in moving an electron through a potential difference of 1 volt is called an "electron volt"

Electron volt (ev),

$$ev = \Delta V \times q$$

$$ev = 1 \times (1.6 \times 10^{-19})$$

$$\boxed{ev = 1.6 \times 10^{-19}\ J}$$

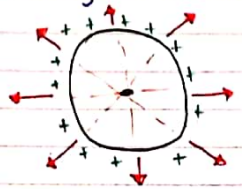
→ electron volt is energy as unit is joule

eg 4.9 Mev

$4.8 \times 10^{6} \times 1.6 \times 10^{-19}$

$\Rightarrow 7.68 \times 10^{-13}\ J$

→ All charge is conductor [illegible] but we assume that charge is concentrated in the centre.

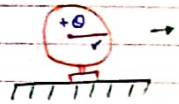


↳ All dotted lines meet at centre so charge is concentrated at centre.

Formulas

$$\boxed{F = \frac{KQ_1Q_2}{r^2}} \quad \boxed{E = \frac{KQ}{r^2}} \quad \boxed{V = \frac{KQ}{r}} \quad \boxed{U = \frac{KQq}{r}}$$

Q. $E = 20\frac{kV}{cm}$ Diameter = 42cm



→ +Q is conc at centre

a) Find charge stored by sphere Q = ?

Ans

$$E = \frac{KQ}{r^2}$$ – electric field strength at surface of sphere

$r = 21cm$ or $0.21m$

$K = 9 \times 10^{9}\ Nm^2/C^2$

$$E = \frac{20kV}{cm}$$

$$\Rightarrow \frac{20 \times 10^{2}}{10^{-2}} = \frac{V}{m} \quad \therefore 20 \times 10^{5}\ Vm^{-1}$$

$$20 \times 10^{5} = \frac{9 \times 10^{9} \times Q}{0.21^2}$$

$$\boxed{Q = 9.8 \times 10^{-6}\ C}$$

b) V = ?

$$V = \frac{KQ}{r}$$

$$V = \frac{9 \times 10^{9} \times 9.8 \times 10^{-6}}{0.21}$$

$$\boxed{V = 4.2 \times 10^{5}\ V}$$

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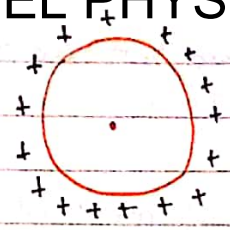
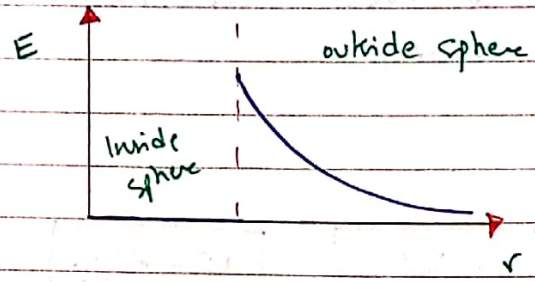
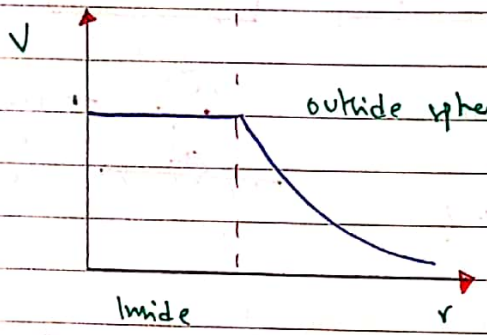
Q





outside sphere → $V = \frac{kQ}{r}$ $V \propto \frac{1}{r}$

potential and distance are inversely proportional so graph would be like this



$E = \frac{\Delta V}{d}$ → Inside charged conductor electric field strength is zero $0 = \frac{\Delta V}{d}$ ∴ $\Delta V = 0$

so potential difference is zero in charged sphere

→ so potential has to be same throughout charged sphere so potential difference will be zero.

(potential 1) − (potential 2)

⇒ 0

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